Technical University of Munich

E0452

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Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
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Diskrete Wahrscheinlichkeitstheorie

Exam:	/ Endterm	**Date:**	Friday 1st August, 2025
Examiner:		**Time:**	08:00 – 10:00

	P 1	P 2	P 3	P 4	P 5
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Working instructions

- This exam consists of **12 pages** with a total of **5 problems**.
  Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 51 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
  - one **single sheet of A4 paper** with handwritten notes on both sides
  - one **analog dictionary** English ↔ native language **without notes**
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.
- Place your student ID on your desk.

Left room from ________ to ________	/	Early submission at ________	

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Problem 1 True or false (12 credits)

Decide whether the statements are true or false (clearly indicate your answer) and provide a justification (proof, counterexample, or explanation via known results).

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a) Let $(\Omega, \mathcal{A}, P)$ be a probability space and $A, B \in \mathcal{A}$ with $P(A) > 0$ and $P(B) > 0$. Then $P(A \mid B) = P(B \mid A)$.

0 1 2 3

b) Let $(\Omega, \mathcal{A}, P_\Theta)$ be a statistical model and let $\hat{\theta}_1$, $\hat{\theta}_2$ be unbiased estimators of θ in P_Θ. Then $\frac{1}{3}\hat{\theta}_1 + \frac{2}{3}\hat{\theta}_2$ is an unbiased estimator of θ as well.

c) Upon receiving a large batch of switches, you take a subsample (uniformly at random) of size n. You run a (suitable) hypothesis test at significance level $\alpha = 0.05$ to test H_0: "the proportion of defective switches in the batch is at most what the manufacturer claimed" versus H_1: "the proportion of defective switches is larger." The test rejects H_0 at level $\alpha = 0.05$. You can therefore conclude that the probability that the entire batch conforms to the claimed proportion is at most 5%.

0 1 2 3

d) Let X, Y be iid following $\mathrm{Exp}(\lambda)$ for $\lambda > 0$. Then the cdf of the ratio X/Y is

$$F_{X/Y}(t) = \begin{cases} \frac{t}{1+t} & \text{if } t > 0\,, \\ 0 & \text{otherwise}\,. \end{cases}$$

Hint: The pdf of the exponential distribution is $p_{\mathrm{Exp}(\lambda)}(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ (and 0 otherwise). Note that $F_{X/Y}(t) = P(X < t \cdot Y)$.

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Problem 2 Definitions (6 credits)

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a) Write out the central limit theorem with all required assumptions.

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b) Give the full general definition of the type of convergence in the central limit theorem.

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Problem 3 Density computations (9 credits)

Let X be a continuous RVRV with probability density function

$$p_X(x) = \begin{cases} \frac{x}{c} & \text{for } x \in [0,2]\,, \\ 0 & \text{otherwise}\,, \end{cases}$$

for some $c \in \mathbb{R}$.

a) Find $c \in \mathbb{R}$ such that p_X is a valid pdf.

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b) Determine the cumulative distribution function (cdf) $F_X(x)$ of X.

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c) Calculate the mean and median of X.

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d) Let $Y = \ln(X)$. Determine the pdf of the random variable Y.
Hint: You may use the change of variables formula for densities: $p_Y(y) = p_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$.

Problem 4 Estimating and testing (12 credits)

Let $X_1, \dots, X_n$ be iid RVRVs from a Poisson distribution $\mathrm{Poi}(\lambda)$ with unknown rate parameter $\lambda > 0$.

a) Explicitly derive the maximum likelihood estimator (mle) $\hat{\lambda}_{\text{mle}}$ for λ.
Hint: The pmf of a $\mathrm{Poi}(\lambda)$ distributed random variable is $p_{\mathrm{Poi}(\lambda)}(x) : \mathbb{N} \to [0,1], x \mapsto \frac{e^{-\lambda}\lambda^x}{x!}$, which has mean and variance λ. Recall that the derivative of $\ln(t)$ is $1/t$ for $t > 0$.

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b) Is $\hat{\lambda}_{\text{mle}}$ an unbiased estimator for λ?

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c) Use the central limit theorem (clt) to find the approximate distribution of $\hat{\lambda}_{\text{mle}}$ for large n.

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d) In a light bulb factory, the number of defectives per batch is assumed to follow a Poisson distribution. Historically, the average number of defects per batch has been $\lambda_0 = 4$. After a change in the manufacturing process, you test $n = 100$ new batches and find an average of $\bar{x}_{100} = 5$ defects.
Construct an *approximate* hypothesis test to test whether the average number of defects per batch has *increased above* the historical average at a significance level of $\alpha = 0.05$. Based on this test, should you reject the null hypothesis?
Hint: You may use that the 0.95-quantile of a standard normal distribution is $z_{0.05} \approx 1.64$.

Problem 5 **Duck hunt (12 credits)**

Let $m, n \in \mathbb{N}_{>0}$. A group of n hunters with perfect aim simultaneously shoot at m ducks, where each hunter selects its victim arbitrarily (that is, uniformly at random) and independent of the other hunters. So it can happen that a duck gets selected by multiple hunters. Let X be the $\mathbb{N}$-valued random variable modeling the number of surviving ducks after all hunters have pulled the trigger (each of them only shoots once).

a) Calculate the expectation $\mathbb{E}[X]$.
Hint: Enumerate the ducks and define the event A_i := "the i-th duck survives." Then construct the random variable X using χ_{A_i}.

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b) Calculate the variance var[X].

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c) Assume now there are 2 hunters. Find $k \in \mathbb{N}$ as a function of m such that the number of surviving ducks is within $[\mathbb{E}[X] - k, \mathbb{E}[X] + k]$ with a probability of at least 90%.

Hint: Rewrite this condition so you can use the Chebyshev inequality.

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Additional space for solutions–clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

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